

# Submission Summary

**Conference Name**

VI International Congress of Interdisciplinary Research #LoHacemosxTIC

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**Paper ID**

21

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**Paper Title**

Development of a Prototype Measurement System for Comparing Atomic Oscillator Signals Using GNSS.

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**Abstract**

Time and frequency transfer using GNSS systems constitutes a fundamental tool for ensuring synchronization and temporal traceability in scientific, industrial, and telecommunications applications. In this context, National Metrology Institutes require measurement systems capable of performing comparisons between atomic oscillators, enabling the generation and maintenance of national UTC(k) time scales, as well as the provision of high-accuracy measurement services.

This work describes the development of a prototype measurement system for the comparison of atomic oscillator signals using the satellite-based Common-View technique with a low-cost GNSS receiver. The project was developed through the collaboration between the groups GICOG and GIMCI, integrating expertise in technological development, data processing, and time and frequency metrology.

The implemented prototype integrates a low-cost GNSS receiver, an external reference provided by an atomic oscillator, and software for GNSS data acquisition and processing. The system enables the capture of raw observations and their conversion into standard formats. Algorithms were developed for observation processing and time correction procedures, allowing the generation of CGGTTS files.

System validation was carried out by comparing the obtained results with those from the reference system developed by NIST, both connected to the same reference. Statistical tools such as error calculation, Type A uncertainty, normalized error  $E_n$ , and correlation analysis were used for the evaluation. The results demonstrated that the developed system satisfies the criterion of maintaining the time scale within  $\pm 100$  ns with respect to UTC, obtaining an error of  $(-3.01 \pm 0.55)$  ns. Furthermore, the compatibility analysis yielded a value of  $E_n = -0.77$ , indicating compatibility between both systems, while the correlation coefficient of 0.98 showed a strong relationship between the prototype results and the reference system.

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## Submission Files

Abstract Prototype Measurement Sysytem.pdf (232.3 Kb, 2026-05-16 19:10:10 COT)

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## Submission Questions Response

### 1. Topics

*Please select a topic to your submission from the following list. Topics are typically used for assigning submissions to reviewers and for conference analytics. You must select exactly one topic.*

Technology and artificial intelligence

### 2. Additional Information

*Please provide the following information for each author listed in this submission: Highest level of education (indicate the name of the degree obtained). Role within the organization (e.g., Student, Professor, Alumni, Researcher). Mobile phone number (including country code).*

*EXAMPLE: Author 1: PhD in Computer Science, Professor, +57 3001234567*

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### 3. I am interested in having the work submitted for publication in a scientific journal

*Select whether you would like your work to be submitted for publication in a scientific journal after review.*

Agreement accepted

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